

Discrete Optimum Design of Geometrically Non-Linear Trusses using Genetic Algorithms

M.N.S. Hadi and K.S. Alvani
Faculty of Engineering
University of Wollongong, Australia

Abstract

This paper presents a formulation of Genetic Algorithm based optimisation for the design of plane and space truss structures with geometrically non-linear behaviour. The cross-sectional areas of truss members are treated as design variables and constraints are imposed on stress, displacement and member instability. Areas of the member are grouped in order to reduce the number of independent design variables and to produce practical designs. An incremental load approach with a Newton-Raphson type of iteration is used in the geometrically non-linear analysis procedure and in developing the optimisation algorithm. A number of design examples are presented to illustrate the application of the proposed algorithm to plane and space truss structures with a large number of design variables. Solving these examples proved that the proposed formulation is viable and leads to optimum solutions.

Keywords: discrete optimisation, genetic algorithms, geometrically non-linear analysis, space trusses, optimum design.

1 Introduction

There are several optimisation techniques that are used to obtain optimum designs of structural systems. Khot and Kamat [1] and Saka [2] independently developed optimisation methods based on optimality criteria technique for the optimum design of geometrically non-linear truss structures with continuous design variables subject to constraints on displacements and system non-linear stability. Saka and Ulker [3] used the same optimisation method for the optimum design of geometrically non-linear space trusses with continuous design variables subject to stress and displacement constraints. It was observed by Saka and Ulker [3] that there could be a significant variation in the solution based on linear and non-linear analyses to the extent that sometimes the direction of the member force is reversed when geometric non-linearity is taken into account indicating its importance in obtaining realistic

solutions. Gutkowski and Zawidzka [4] used sequential and enumeration methods for discrete minimum weight of trusses subject to stress, displacement and cross-sectional area constraints.

In recent years, genetic algorithms (GAs), which are applications of biological principle into computational algorithms, has become most successful and powerful for solving structural design problems, because of their simplicity, global perspective and inherent parallel processing. Detailed and formal proof for this optimisation technique can be found in [5]. Many research studies have recently reported the solution of truss structure optimisation problems via GAs. Golberg and Samtani [6] appear to be the first to suggest the use of GAs for the optimal design of truss structure. Rajeev and Krishnamoorthy [7], Shyue and Pei-Tse [8] and Erbature et al. [9] used GAs to find the minimum weight design of trusses with discrete design variables. Coello and Christiansen [10] presented the optimum design of trusses based on multiobjective structural optimisation using GAs. GAs have also been used to determine the best structural topology as illustrated by Hajela and Lee [11], Ohsaki [12] and Grieson and Pak [13]. Rajan [14] used the GA to design truss structures by combining sizing, shape and topology.

All these studies have shown that GA is an effective tool for the optimal design of truss structures and is simple to implement when compared to other optimisation techniques. Furthermore, consideration of design variables as discrete quantities is essential in the optimisation of most structural system.

Much of the reported optimisation techniques for the optimum design of trusses are based on linear behaviour. However, in structural optimisation there are cases, such as long-span and slender structures (for example suspension bridges), where non-linear analysis is necessary to better represent the behaviour of the structures under different loading conditions. In addition, much of the optimisation techniques used for the optimum design of structures are based on continuous variables, thus yielding member dimensions that are practically unachievable. None of the previous studies is based on non-linear analysis with discrete variables using GA as the optimisation tool.

In the present study an optimum design algorithm is developed for geometrically non-linear plane and space trusses composed of elements that are chosen from a given set of cross-sections. In the optimisation process the non-linear analysis of trusses are solved by an incremental strategy, non-linear equation are linearised and iteratively solved by the Newton-Raphson method.

In order to bench mark the solutions obtained using non-linear analysis, an optimisation technique based on linear analysis is also developed. Results from the optimisation technique based on non-linear analysis are compared with the linear analysis. In addition, when possible these results are compared with the results of published examples. These comparisons proved that the developed technique is a viable technique that leads to optimum structures efficiently.

2 Genetic Algorithms

GAs are stochastic search techniques based on the principles of survival of the fittest and adaptation. According to Goldberg [5] the application of genetic algorithms for

adaptive systems was first proposed by J. H. Holland in 1962. The GA is a population-based algorithm. The population may consist of several potential solutions, which are called individuals. These individuals are subjected to different genetic operations such as selection, crossover and mutation. These operations are designed to drive the whole population to a near global optimal solution. GAs do not require gradient or derivative information. For this reason alone, it has been suitable to be used for hard and complex optimisation problems and is capable of obtaining near optimum solutions in a simple way. GA differs from traditional gradient-based optimisation techniques in terms of the following: they work with a coding of the design variables rather than the variables themselves, and they search a population of possible solutions instead of a single point. GAs use stochastic reproduction instead of deterministic rules.

3 Optimum Design Problem

The optimum design problem of a geometrically non-linear truss structure with n elements, m degrees of freedom and discrete member size can be expressed as follows:

Minimise the weight

$$W(A) = \sum_{i=1}^n A_i L_i \rho \quad (1)$$

subject to

$$\begin{aligned} |\sigma_i| &\leq \sigma_i^a & i=1, \dots, n \\ |d_j| &\leq d_j^a & j=1, \dots, m \\ A_i^l &\leq A_i \leq A_i^u \\ A_i &\in AC \end{aligned} \quad (2)$$

where $A = \{A_1, A_2, \dots, A_n\}^T$ is an n -dimensional vector of design variables, L_i is the length of the i^{th} member, ρ is the weight density of material and, σ_i is the stress in the i^{th} member, σ_i^a is the allowable stress in both tension and compression, d_j is the displacement of the j^{th} degree of freedom, d_j^a is the allowable displacement, AC is a set of available cross-sections area and the superscripts l and u reflect the lower and upper bound values of design variables, respectively.

Solving the above problem will result in a discrete optimum design for the truss structure problem.

In the case of dominant buckling constraints in the design problem, the stress constraint is defined as:

$$|\sigma_i| \leq \sigma_i^{+a} \text{ or } |\sigma_i| \leq \sigma_i^{-a}, \quad i=1, \dots, n \quad (3)$$

where σ_i^{+a} and σ_i^{-a} are the allowable stresses in tension and compression, respectively. According to the AISC specification σ_i^{+a} is given as $0.6F_y$, where F_y is the yield stress of steel and the allowable compressive stress σ_i^{-a} is defined as:

if $S_i < C$ then:

$$\sigma_i^{-a} = \left[\frac{1 - \frac{S_i^2}{2C^2}}{\frac{5}{3} + \frac{3S_i}{8C} - \frac{S_i^3}{8C^3}} \right] \times F_y \quad (4)$$

else, if $S_i \geq C$ then:

$$\sigma_i^{-a} = \frac{12\pi^2 E}{23S_i^2} \quad (5)$$

where S_i is the slenderness ratio for i^{th} member and C is given as $\sqrt{2\pi^2 E/F_y}$ where E is modulus of elasticity. Details of the formulas can be obtained from Ref. [15].

GAs were developed to tackle unconstrained optimisation problems. However, constrained optimisation problems are solved by transforming the problems into an unconstrained problem by including a penalty function as suggested in [7]. The unconstrained optimisation problem is then written as:

Minimise

$$U(A) = W(A) \times P \quad (6)$$

where $U(A)$ is the unconstrained objective or fitness function, P is a penalty value, if no violation is found, then all penalty values is 1, *i.e.* no penalty is imposed on the objective function. If a constraint is violated, then the penalty is defined as:

$$P = (1 + dc \times (Cs + Cd)) \quad (7)$$

where dc is known as a violation parameter, and is selected depending on the required influence of a violated individual in the next generation, a value of 30 was found suitable for all the design examples presented in this study, Cs and Cd are stress and displacement violation coefficients, respectively, which are defined as:

$$Cs = \sum_{s=1}^n gs \quad \text{and} \quad Cd = \sum_{d=1}^m gd \quad (8)$$

where

$$\begin{aligned} gs &= |\sigma_i| / \sigma_i^a - 1 \quad \text{if} \quad |\sigma_i| / \sigma_i^a - 1 > 0 \\ gd &= |d_j| / \sigma_j^a - 1 \quad \text{if} \quad |d_j| / \sigma_j^a - 1 > 0 \end{aligned} \quad (9)$$

and the number of constraints is $(n+m)$.

In the case of dominant buckling constraints in the design problem, the value of gs is defined as:

$$gs = g^{-a} + g^{+a}, \quad \begin{cases} g^{-a} = \frac{|\sigma_i^-|}{|\sigma_i^{-a}|} - 1 & \text{if } \frac{|\sigma_i^-|}{|\sigma_i^{-a}|} - 1 > 0 \\ g^{+a} = \frac{\sigma_i^+}{\sigma_i^{+a}} - 1 & \text{if } \frac{\sigma_i^+}{\sigma_i^{+a}} - 1 > 0 \end{cases} \quad (10)$$

where σ_i^- and σ_i^+ are respectively, stresses in compression and tension in the i^{th} member, σ_i^{-a} and σ_i^{+a} are the allowable stresses in compression and tension, respectively.

It can be easily seen that if a truss member violates a constraint, a penalty parameter is added to the weight, $W(A)$, making the truss less desirable because its weight has been artificially increased. This penalised weight is defined as individual's fitness $U(A)$ as expressed in equation (6). These fitness values are the usual starting point for implementation of a GA in an optimisation problem. In the first operator in GAs, the reproduction operator, a mating pool is created by letting individuals with higher fitness values have more chance to be selected and allowing them to recombine, i.e., crossover and mutation. In order to obtain the minimal fitness $U(A)$, the algorithm requires a condition to perform the selection between the individuals. This is done in such a way, that the best individual has the highest fitness. An expression for the fitness becomes:

$$\Phi_i = [U(A)_{\max} + U(A)_{\min}] - U_i(A) \quad (11)$$

where Φ_i is the fitness of the i^{th} individual, $U(A)_{\max}$ and $U(A)_{\min}$ are respectively the maximum and minimum values of unconstrained function of Eq. (6).

After the evaluation of fitness value, the fitness factor (probability of selection) Fc for the i^{th} individual is calculated by:

$$Fc(I_i) = \frac{\Phi(I_i)}{\sum_{j=1}^n \Phi(I_j)} \quad (12)$$

where I_i represents the i^{th} individual in the population and n is the population size.

According to the fitness factor individuals get a number of copies in the mating pool. Highly fit individuals get more copies in the mating pool, whereas the less fit ones get fewer copies.

4 Geometrically Non-linear Analysis

Geometric non-linearity occurs when the deflections are large enough to cause significant changes in the geometry of the structures. Although the materials that constitute a truss behave in a linearly elastic manner, the overall external load deflection relationship of these members may become non-linear. It is therefore the equations of force equilibrium of the structure that must be established in the current configuration. However, because of large deflection, the strain-displacement equations contain non-linear terms, which must be included in calculating the stiffness matrix. The non-linear effects in the strain-displacement equations modify the overall stiffness matrix:

$$[K_T] = [K_E] + [K_G] \quad (13)$$

where $[K_T]$ is known as total tangent stiffness matrix in global coordinates of structure, $[K_E]$ is linear elastic stiffness matrix and $[K_G]$ is the geometric stiffness matrix in the global coordinate.

Thus it can be seen from equation (13) that the total stiffness of a truss element consists of two parts, the elastic stiffness matrix $[K_E]$, which is the same as that used in linear analysis and the geometric stiffness matrix $[K_G]$, which describes the changes in geometry of the element. During loading the structure, initially no change in geometry is registered, $K_G = 0$, rendering this stage similar to a linear analysis. As the loading is increased, members of the truss experience a change in geometry, which then requires non-linear analysis.

4.1 Geometric Stiffness Matrix for Plane Truss

The derivation and details of the formulas and more explanation for the analysis can be obtained from [16-17]. The geometric stiffness matrix K_G of an element in the global coordinate is given as:

$$K_G = \frac{P}{L} \begin{bmatrix} \mu^2 & -\lambda\mu & -\mu^2 & \lambda\mu \\ -\lambda\mu & \lambda^2 & \lambda\mu & -\lambda^2 \\ -\mu^2 & \lambda\mu & \mu^2 & -\lambda\mu \\ \lambda\mu & -\lambda^2 & -\lambda\mu & \lambda^2 \end{bmatrix} \quad (14)$$

where P is the intermediate axial load of the truss element at the current loading stage, L is the element length. $\lambda = \cos \phi$ and $\mu = \sin \phi$, (ϕ is the angle of orientation of the member).

4.2 Geometric Stiffness Matrix for Space Truss

The geometric stiffness matrix K_G of an element is given in [18]:

$$(K_G)_i = \frac{P_i}{L_i} \begin{bmatrix} (K_G)_i^B & -(K_G)_i^B \\ -(K_G)_i^B & (K_G)_i^B \end{bmatrix} \quad (15)$$

where P_i is the force in member i , L_i is its length and $(K_G)_i^B$ is the transformation matrix given by:

$$(K_G)_i^B = \begin{bmatrix} 1-\lambda^2 & -\lambda\mu & -\lambda\nu \\ -\lambda\mu & 1-\mu^2 & -\mu\nu \\ -\lambda\nu & -\mu\nu & 1-\nu^2 \end{bmatrix} \quad (16)$$

where:

$$\begin{aligned} \lambda &= \cos(\phi_{xx}^-), \phi_{xx}^- \text{ is angle between } \bar{x}, x \\ \mu &= \cos(\phi_{yx}^-), \phi_{yx}^- \text{ is angle between } \bar{y}, x \\ \nu &= \cos(\phi_{zx}^-), \phi_{zx}^- \text{ is angle between } \bar{z}, x \\ &\vdots \\ &\vdots \\ \nu &= \cos(\phi_{zz}^-), \phi_{zz}^- \text{ is angle between } \bar{z}, z \end{aligned}$$

The global equilibrium equations for both plane and space truss structures are

$$[K_L]\{d\} = \{F\} \quad (17)$$

where $\{d\}$ is the vector of nodal displacements, $\{F\}$ is the vector of external applied loads at the nodal points.

Since the global stiffness matrix $[K_L]$ depends on member forces and angle of member orientation ϕ , which are both functions of nodal displacements. Thus the equilibrium equation (17) represents a system of non-linear equations and an iterative solution scheme will be required to solve these non-linear equations.

4.3 Computational Technique

In the incremental load methods, the Newton-Raphson method is employed on the basis of iterative procedures and applied to the above geometrically non-linear analysis problems. The Newton-Raphson method begins at a known point i on the equilibrium point. An increment load ΔF_i , of the total external load is applied to the structure, then the incremental displacements $\{\Delta u\}$ is calculated by making use of

the system tangent stiffness matrix $[K_t]_i$ for the load increment i and solving the global system of equations:

$$[K_t]_i \{\Delta u\} = \{\Delta F\}_I \quad (18)$$

where $\{\Delta F\}$ and $\{\Delta u\}$ are incremental values of external loads and joint displacements, respectively.

The increment of displacements is added to the previous displacement to form a new total displacement $\{u_i\} + \{\Delta u\}$. The new displacement is not necessarily the correct value of the displacement of structure at a load level $\{F\}_{i+1}$. Therefore the joint equilibrium equations are not satisfied, and indicate that the internal nodal forces are not in equilibrium with the nodal external forces. This difference of displacement can be improved by applying a Newton-Raphson type of iteration.

5 Optimum Design Algorithm

After defining the GA parameters, which include population size, maximum generation, number of design variables, crossover probability, mutation probability, upper and lower boundaries of the design variables and the fitness function, the GA starts by initialising a set of binary strings (initial population). The initial population is usually created randomly and represents a possible solution within the domain of the solution space. For each individual the binary design variables are decoded into decimal values and the sequence numbers in the available section list are obtained. Then a geometrically non-linear analysis procedure is performed. The member stresses and nodal displacements results at the end of the non-linear analysis are used to evaluate the fitness function value for each individual and based on the fitness factor individuals are selected and placed into the mating pool. New individuals for the next generation are created through crossover and mutation process. This process continues until the optimum solution (minimum total weight of truss) is obtained.

In each optimisation cycle the structure is analysed using the geometrically non-linear procedure. An elitist strategy is also used, that mutates the best string before it is copied to the next generation. Then the objective value of the mutated string is compared with the original one. If the mutated string is better, it replaces the original one (and is copied to the next generation). Otherwise, the original string is copied to the next generation.

The optimum design formulation presented in this study was written in Matlab and a number of examples were solved.

6 Design Examples

To investigate the feasibility of the developed algorithm, a 10-bar plane truss, a 20-bar plane truss, a 51-bar roof trusses, a 56-bar space truss and a 120-bar space truss were solved and results are presented. The GA is based on a roulette-wheel reproduction scheme [5], a one-point crossover, and a standard mutation scheme. In

all examples the values of 0.8 and 0.02 are used for the probability of crossover and mutation, respectively. The population size and number of generations (ng) are fixed after trial runs. The length of the bit string depends on the required precision and size of the search domain namely lower and upper boundaries, and performed by using:

$$2^{(l_i-1)} < (D_{(max)_i} - D_{(min)_i}) \times 10^{K_i} \leq 2^{l_i} \quad (19)$$

where l is the length of binary string, $D_{(max)}$ and $D_{(min)}$ are the upper and lower bound of the search domains respectively; K is the required decimal precision and the subscript i denotes the i^{th} design variable.

6.1 Design of 10-bar Plane Truss

The first problem to be considered is the 10-bar truss taken from Ref. [4] shown in Figure 1 where it was solved assuming (1) continuous optimisation problem and (2) discrete optimisation problem. Details of the dimensions and loading considered are presented in Figure 1. The total weight of the truss structure is considered as the objective function, which is solved using geometrically non-linear analysis under the constraints of member stresses and nodal displacements. The buckling constraint was not dominant in this design problem.

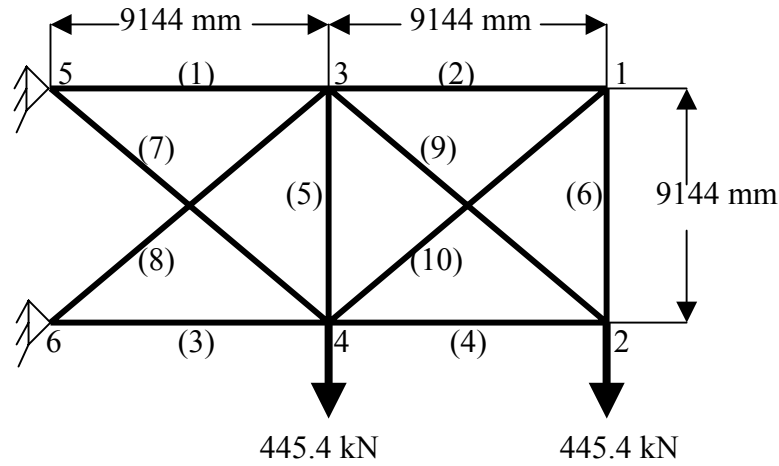


Figure 1: Geometry and loading of 10-bar truss

The problem was solved for the following values of structural parameters: the modulus of elasticity was 6.89×10^4 MPa, and the material density 2770 kg/m^3 the allowable displacement was limited to 50.8 mm and the allowable stress to 172.25 MPa in both tension and compression for all members. The problem has 10 design variables. In the case of discrete optimisation the design variables were selected from the following set of discrete values of two catalogues. The first catalogue was:

(65, 323, 645, 1290, 2581, 4516, 7742, 12258, 17419, 23226) (mm^2), and the second catalogue was: (65, 645, 1290, 3226, 5161, 7742, 9677, 11613, 12903, 16129, 19355, 22581, 29032) (mm^2). In the case of continuous optimisation the lower and upper bound of cross-sectional areas were given as 65mm^2 and 29032mm^2 , respectively.

Case	Design Variables	Cross-section areas (mm^2)				
		Gutkowski-Zawidzka [4]			Proposed method	
		Continuous Solution	Sequential Algorithm	Enumeration Algorithm	Continuous Solution	Discrete Solution
Case 1*	A1	19375	23226	23226	19652	17419
	A2	65	326	323	65	65
	A3	15015	17419	17419	13329	17419
	A4	9862	12258	12258	11690	12258
	A5	65	323	323	65	65
	A6	365	323	1290	65	65
	A7	4818	4516	4516	5050	4516
	A8	13676	17419	12258	13643	12258
	A9	13947	12258	12258	13458	17419
	A10	66	323	65	65	65
	Weight (kg)	2295.9	2484.7	2429.5	2288.7	2423.2
Case 2†	A1	19375	19355	19355	19652	19355
	A2	65	65	65	65	65
	A3	15016	19355	16129	13329	16129
	A4	9862	9677	7742	11690	7742
	A5	65	65	65	65	65
	A6	365	65	645	65	65
	A7	4818	5161	5161	5050	5161
	A8	13676	12903	12903	13643	16129
	A9	13947	12903	16129	13458	12903
	A10	65	65	65	65	65
	Weight (kg)	2295.9	2340.4	2339.9	2288.7	2325.2

*Case 1 was based on the solution from catalogue 1.

†Case 2 was based on the solution from catalogue 2.

Table 1: Optimal Design of 10-bar Plane Truss

The final results of the optimal design problem are listed in Table 1 with the comparison with results obtained by Gutkowski and Zawidzka [4]. The results of optimisation using the first catalogue are presented as case 1 and the results of optimisation using second catalogue are presented as case 2. Comparing the results of both continuous and discrete optimisation solutions shows that the present algorithm gives a better minimum weight.

2.2 Design of 20-bar Plane Truss

The geometry and nodal coordinates of this example are shown in Figure 2. The structure is subjected to two load conditions of 850 kN each at nodes 4 and 6. The objective is to minimize the weight of the structure by considering discrete design variables and geometrically non-linear analysis. The modulus of elasticity is specified as 210×10^3 MPa, and material density as 7850 kg/m^3 . The allowable displacement is limited to 50 mm and the yield stress according to Ref. [19] for the circular tubeline steel used is 350 MPa. The solution of the problem is obtained in two stages. The first stage is based on the stress and displacement constraints and in the second stage the member buckling constraints are included in the design problem. In the case of dominant buckling constraints, all member stresses are constrained to be below the buckling stress.

In the optimisation process the design variables are selected from the following set of available sections: $A \text{ (mm}^2\text{)} = (80.9, 96.6, 108, 121, 130, 153, 156, 178, 182, 199, 254, 325, 332, 414, 419, 453, 523, 533, 705, 733, 809, 862, 989, 1110, 1120, 1230, 1240, 1250, 1270, 1290, 1400, 1440, 1500, 1530, 1550, 1650, 1660, 1780, 1830, 1860, 1910, 2040, 2280, 2230, 2470, 2700, 2760, 3230, 3540, 3600, 4020, 4050, 4280, 4420, 4670, 5360, 5430, 5440, 6380, 7020, 7710, 8040, 8230, 9060, 9380)$. In total there are 65 different available sections. This list of sections area is taken from Ref. [19]

In order to simplify the supply of members and the fabrication of the structure, it was decided to categorise the members into groups. Furthermore, grouping of members has a large effect upon the optimum design problem in that it greatly reduces the number of design variables in the problem. Thus the 20 members of the structure are divided into ten groups, which are required to have the same cross-section area. The final solution has been obtained using a population size of 30 and six iterations were required for convergence in the non-linear analysis. The grouping of members and the results of the optimal design problem are presented in Table 2.

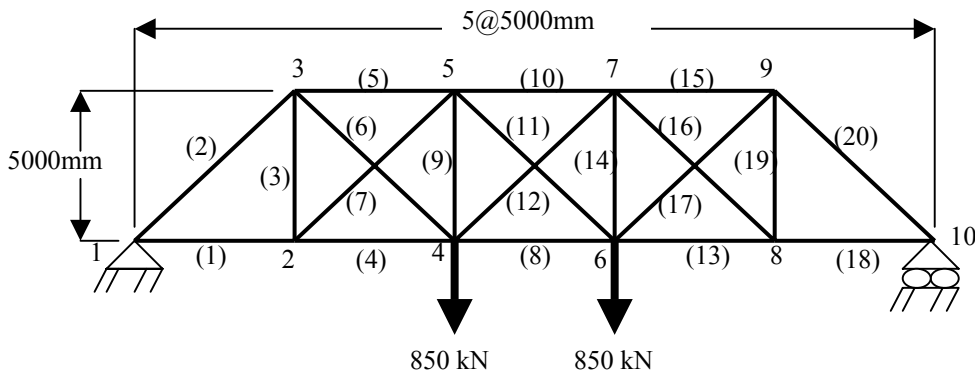


Figure 2: Geometry and loading of 20-bar truss

Group	Member	Cross-section area (mm ²)	
		Case 1*	Case 2†
A1	1, 4, 8, 13, 18	10400	10400
A2	3, 9, 14, 19	1860	2230
A3	5, 10, 15	12400	11800
A4	2, 20	10400	10300
A5	6	8040	8040
A6	7	2700	2470
A7	11	1110	733
A8	12	1780	1650
A9	16	1290	6380
A10	17	8040	4670
Total weight (kg)		6222	6253

*Case 1 discrete non-linear solution.

†Case 2 discrete linear solution

Table 2: Member Grouping Details and Results of Optimisation of 20-bar plane truss

6.3 Design of 51-bar Roof Truss

This example has been solved by Saka [20] by adopting double angle section for all members. It was treated as continuous optimisation problem based on linear analysis. Saka [20] noted that the buckling constraints were dominant in the design problem. The 51 members of the roof truss are divided into four design variables. The dimensions, details of the members grouping and two load cases considered in the design problem are shown in Figure 3. The slope of the upper chord is taken as 5°.

The objective of this problem is to minimise the total volume of the structure by considering the following cases: (1) continuous design variables subject to stress, displacement and member buckling constraints, (2) discrete design variables subject to stress and displacement constraints, (3) discrete design variables subject to stress, displacement and member buckling constraints.

In the first case the design variables were considered as continuous. In this case the material was assumed to be double angle steel sections with minimum size of cross-section area of 200 mm². The allowable yield stress was taken as 275 MPa and the radius of gyration R is calculated based on the cross-sectional area (A) of the member using $R=0.584A^{0.524}$, [20]. In the second and third cases the values of design variables were considered as discrete. Available sections considered for both cases are the same as those used in the second design example above. The allowable yield stress was taken as 350 MPa. In the second case the buckling constraint is not dominant in design. Results for the three different cases in which the allowable displacement are given as 50 mm and the modulus of elasticity is specified as

210×10³ MPa are summarised in Table 3 and are compared with the results given in Ref. [20].

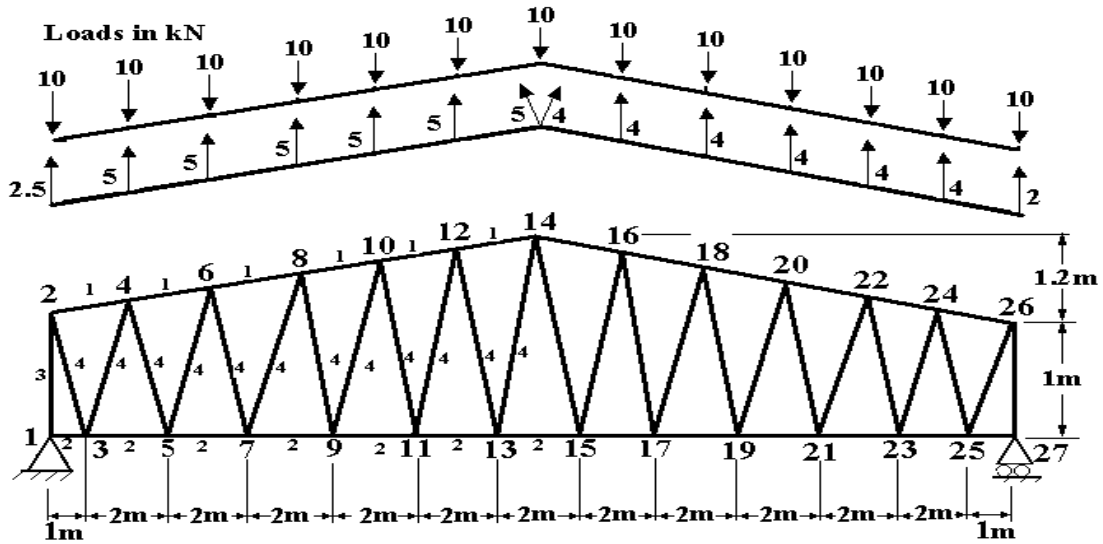


Figure 3: Geometry and loading of 51-bar roof truss

Group	Cross-section area (mm ²)			
	Saka [20]	Proposed method		
		Case 1*	Case 2†	Case 3‡
A1	1673	1473	1500	1230
A2	1149	1095	1240	1500
A3	378	1550	523	1120
A4	721	665	332	533
Volume (x10 ³ mm ³)	101537	95172	82098	92169

*Case 1 is the continuous solution based on member buckling and displacement constraints.

†Case 2 is the discrete solution based on stress and displacement constraints.

‡Case 3 is the discrete solution based on stress, displacement and member buckling constraints.

Table 3: Optimum Design of 51-bar Roof Truss

6.4 Design of 56-bar Space Truss

Consider the 56-bar space truss taken from Saka and Ulker [3] shown in Fig. 4. It was treated as a continuous optimisation problem and designed based on linear and

non-linear analysis. The weight of the structure was taken as the objective function with constraints imposed on the member stress and nodal displacements. The 56-members of the space truss structure are divided into four groups, and the same cross-section area is assumed for each group. Details of the dimension and grouping of members are presented in Figure 4. The space truss is designed under the external loading of 45.5 kN in the horizontal x-direction at joints 1, 2, 5, 7, and 9. 11. 13 and 15 and 91 kN in the negative z-direction at joints 1, 2, 3 and 4. The modulus of elasticity is specified as 210,000 MPa and material density as 7850 kg/m³. The allowable displacements are limited to 30 mm in the x-direction.

The space truss was designed by considering the following cases: (1) continuous design variables using non-linear analysis, (2) discrete design variables using linear analysis, and (3) discrete design variables using non-linear analysis.

Group	Cross-section area (mm ²)				
	Saka and Ulker [3]		Proposed method		
	Non-linear	Linear	Case 1*	Case 2†	Case 3‡
A1	1136	744	791.6	809	705
A2	10878	11102	10760.8	10400	11800
A3	927	500	500	80.9	80.9
A4	4562	4646	4958.4	5360	4280
Weight (kg)	13723.37	13577.16	13642	13590	13569

*Case 1 is the continuous solution based on non-linear analysis.

†Case 2 is the discrete solution based on linear analysis.

‡Case 3 is the discrete solution based on non-linear analysis.

Table 4: Optimum Design of 56-bar Space Truss

In the case of continuous optimisation the allowable yield stress was taken as 240 MPa and the minimum size of cross-sectional area was taken as 500 mm². In the case of discrete optimisation the available sections considered are the same as those used in the second design example above and the allowable yield stress was taken as 350 MPa.

The final results of the optimum design problem for the above three different cases and the results obtained by Saka and Ulker [3] are compared and listed in Table 4. As can be seen from these results, a better minimum weight is obtained by the proposed method when compared with the results obtained by Saka and Ulker [3] using the optimality criteria approach with considering continuous design variables. It is also observed that by including the geometrically non-linear analysis in the design process a better solution is obtained.

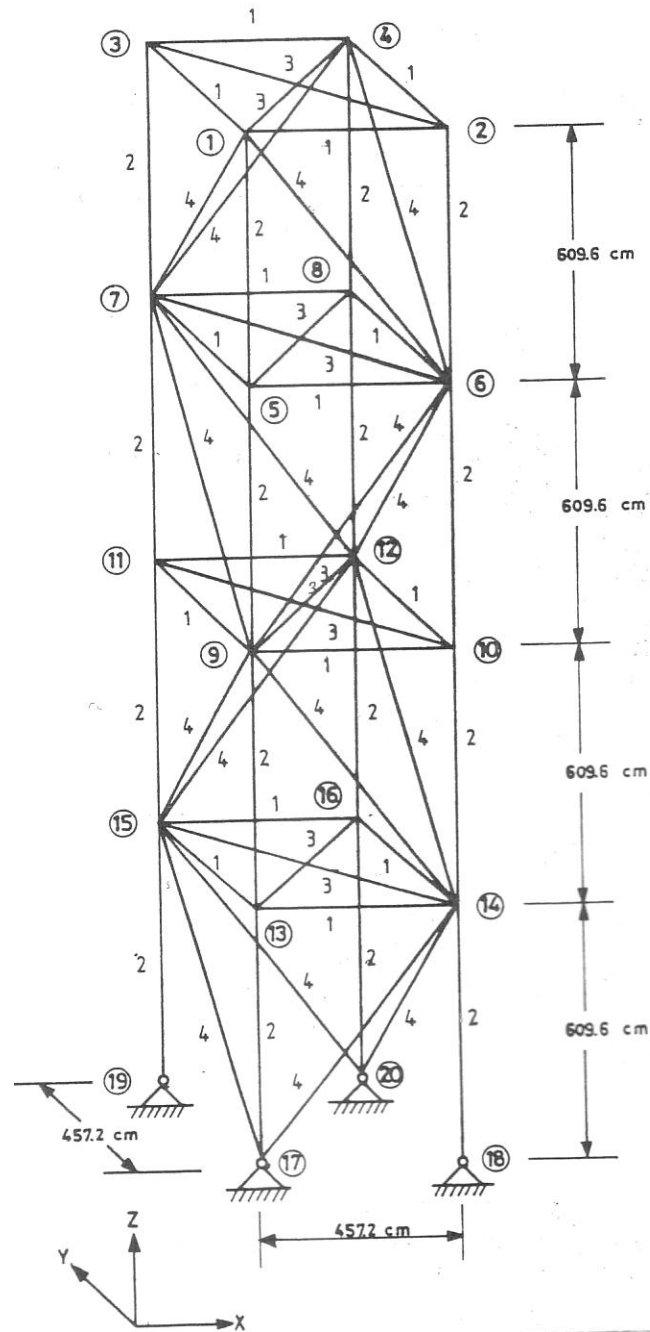


Figure 4: 56-bar space truss [3]

6.5 Design of 120-bar Space Truss

The last problem to be considered is the 120-bar space truss as shown in Figure 5. This problem was also designed as a continuous optimisation problem based on both linear and non-linear analysis [3]. The weight of the structure was taken as the objective function with constraints imposed on the member stress and nodal displacements. The 120 members of the space truss structure are divided into seven different groups and the same cross-sectional area is assumed for each group. The details of the dimension and grouping of members are presented in Figure 5. The space truss is subjected to a vertical load of 60 kN at joint 1, 30 kN at joints 2-14 and 10 kN at joints 15-37, acting in the negative direction of the vertical z-axis. The modulus of elasticity is specified as 210,00 MPa and material density as 7850 kg/m³. The allowable displacements are limited to 30mm in the horizontal x-direction.

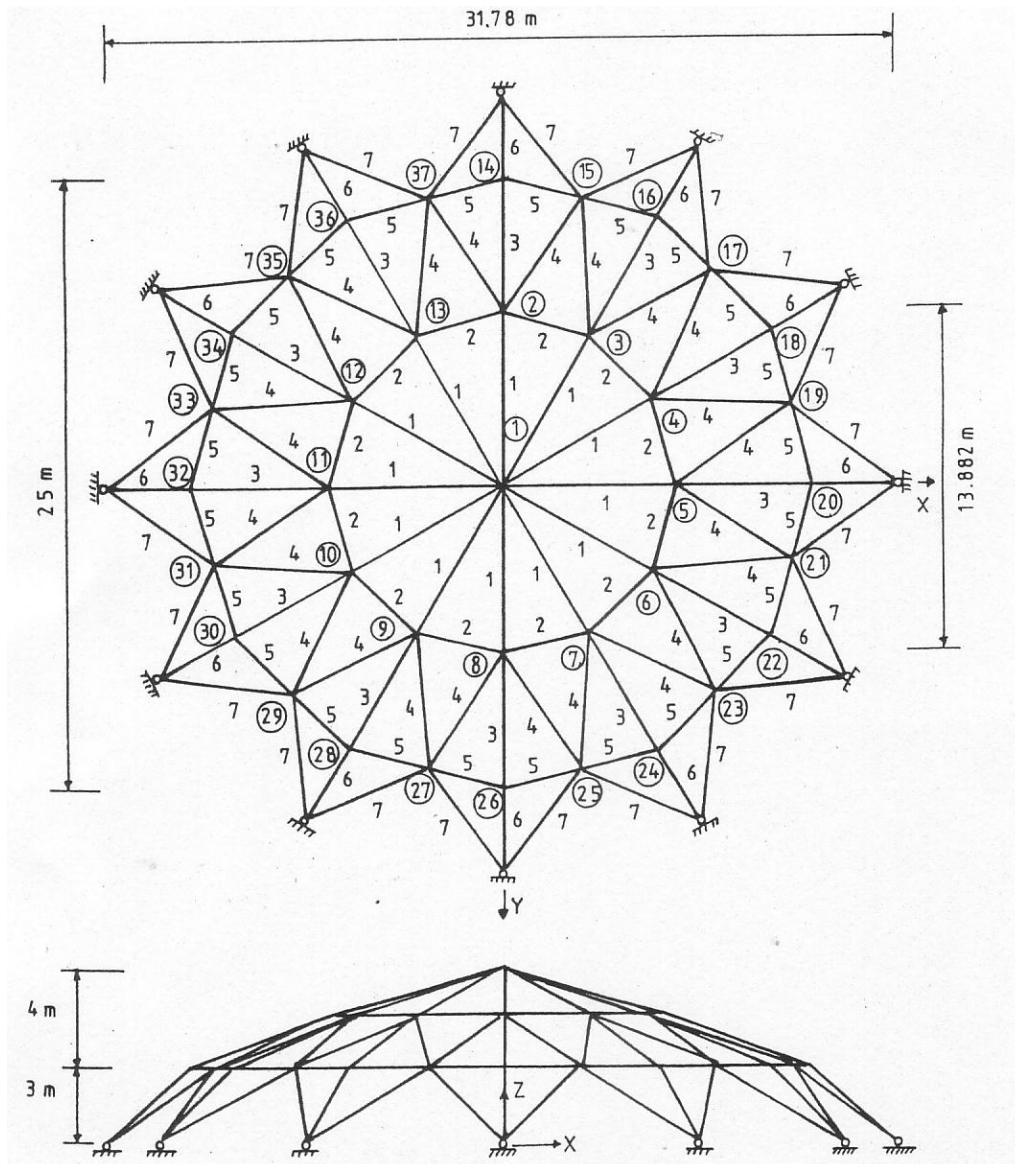


Figure 5: 120-bar space truss [3]

The space truss was designed by considering the following cases: (1) continuous design variables using non-linear analysis, (2) discrete design variables using linear analysis, (3) discrete design variables using non-linear analysis, (4) discrete design variables using non-linear analysis. In this case member buckling constraints are included in the design problem.

In the case of continuous optimisation the allowable yield stress was taken as 240 MPa and the minimum size of cross-sectional area was taken as 200 mm². In the case of discrete optimisation the available sections considered are the same as those used in the second design example above and the allowable yield stress was taken as 350 MPa.

Results from the four different cases are presented together with the results obtained by Saka and Ulker [3] in Table 5. The solution of the current algorithm in case 1 is 7158.6 kg and the solution obtained by the optimality criteria approach [3] is 7587 kg. Here, comparing with the results of continuous optimisation solutions shows that the GAs-based approach gives a better minimum weight. Furthermore, it is observed that further reduction is possible in the weight of the space truss considering the geometrically non-linear analysis as compared to linear analysis.

Method	Analysis	Cross-section area (mm ²)							Weight (kg)
		A1	A2	A3	A4	A5	A6	A7	
Proposed method	Case 1*	1085	3870	3540	523	2737	1530	200	7158.6
	Case 2†	1500	4670	2700	705	2760	1110	182	7264.6
	Case 3‡	1230	5360	2760	533	2470	1780	153	7229
	Case 4§	1650	4420	2470	809	3230	2230	809	8750.7
Saka & Ulker [3]	Non Linear	1750	4556	2545	844	2230	1596	390	7587
	Linear	1666	4489	2489	966	2193	1659	1174	8511

*Case 1 is the continuous solution based on non-linear analysis.

†Case 2 is the discrete solution based on linear analysis.

‡Case 3 is the discrete solution based on non-linear analysis.

§Case 4 is the discrete solution based on non-linear analysis subject to stress, displacement and buckling constraints.

Table 5: Optimum Design of 120-bar Space Truss

7 Conclusions

A GA based optimisation method is presented in this paper for discrete optimal design of plane and space truss structures based on geometrically non-linear analysis. It is found that GA is effective and reliable to solve discrete optimisation problems and can be applied to any discrete set of sections produced according to the different standards. Furthermore it is possible to obtain realistic solutions by

including the geometric non-linear analysis in the design of trusses. Most of the design results are compared with the best solution reported in the literature. In all design problems, the results obtained by the proposed algorithm are better illustrating the advantages of the proposed GA-based methodology for the discrete and continuous truss structural optimisation.

The efficiency of the structural design optimisation is often measured by the GA parameters such as crossover and mutation probability, population size and number of generations. However, because of the random nature of the search, the guidelines are extremely difficult to establish the GA parameters. Generally, it is recommended to have a high crossover and a low mutation probability [5]. The population size and the maximum number of generations were fixed after trying various values and studying the convergence history of the algorithm. Generally, the bigger the population size, the more design features are included.

It was noticed from solving the numerical design problems that most of the computation time was used by the non-linear analysis routine algorithm. It was also noticed that the computing time increases depending on the population size and the number of design variables, a large population size implies longer waiting times for convergence because in each cycle of optimisation process, the non-linear analysis requires 3-15 or more iterations.

Based on the results obtained from this study, it is clear that the developed GA-based optimisation technique is viable and leads to results that are both optimal and satisfy all the constraints of stress, buckling and deflection.

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